YEAR 7 · ICE ENGINEERS 冰雪工程师 · WEEK 3 OF 10 · 40 MINUTES

Meet the variables 认识变量

In a fair test we change only one thing, measure one thing, and keep everything else the same.

- I can name the independent, dependent and control variables in an experiment.
- I can explain why a fair test changes only one thing.

variable 变量 **independent variable 自变量** **dependent variable 因变量** **control variable 控制变量** **fair test 公平测试** **air resistance 空气阻力** **spinner 纸直升机** **wing length 翼长**

THE LESSON

Do Now	4	Two paper planes picture: 'Is this a fair race?' answered with the frame; Today slide.
Hook	3	Teacher drops 4 cm and 8 cm spinners together from 2 m; class votes which lands first.
Learn	7	'I change / I measure / I keep the same' with gestures and vocab panel; cloze; safety slide.
Do	18	Make three spinners (4, 6, 8 cm), drop from 2 m, time, record; picture sort, variables table.
Think and Reflect	8	Odd one out: which test is not fair? Pair talk, share with a frame; exit ticket.

Hands-on: Groups cut and fold three card spinners (4, 6, 8 cm wings), drop each from a 2 m wall mark and time the fall.

Collect or assess: Variables table (Core D) and results (Core C); exit ticket: IV, DV and one control variable, collected at the door.

SAFETY

- Falls: no standing on chairs or tables; Dropper stands flat (1.8 m mark if needed).
- Scissors points down, cut away from fingers; never unbend or point paper clips.
- Tripping: keep bags off the floor under the drop marks.

WATCH OUT

- Check drop height as groups drift lower; 1.8 m mark only if no one reaches 2 m, and then for every drop.
- IV and DV mixed up, or DV called the thing kept the same: go back to the hand gestures.

Language: Frames: 'It is / is not fair because...', 'I change, I measure, I keep the same'. Chinese welcome; Band 1 beside buddy.

Next week: Keep spinners in labelled envelopes and leave wall marks up: Week 4 adds repeats, means and anomalies.

PREP FOR ONE CLASS

before the lesson about 57 min (45 could be done by someone else); after about 20 min (some times estimated)

THE WEEK BEFORE

- [ ] Send worksheet to reprographics: 24 copies, 72 sheets, double-sided, B&W 5 min · reprographics
- [ ] Print 9 spinner sheets on 160 gsm card at 100%; check the 1 cm square 5 min · technician
- [ ] Print 6 odd-one-out sheets (single-sided), cut each into A6 cards A to D 10 min · technician
- [ ] Print one kit card (A5) for the teacher 2 min · teacher
- [ ] Tape 2.0 m and 1.8 m wall marks at 6 stations, away from windows and vents 10 min · technician
- [ ] Make demo S (4 cm) and L (8 cm) spinners, one clip each; test from 2 m 10 min · teacher

THE DAY BEFORE

- [ ] Label six A5 envelopes with the group names 5 min · technician

ON THE DAY, BEFORE THE LESSON

- [ ] Set out 6 kit trays and teacher desk; check wall marks and clear floor 10 min · technician

AFTER THE LESSON

- [ ] Check envelopes hold S, M, L spinners; store with logbooks in class box 5 min · teacher
- [ ] Count back scissors and stopwatches, collect card sets, return kit, tidy 5 min · technician
- [ ] Read exit tickets to plan Week 4 Do Now recap and spot who is Stretch-ready 10 min · teacher

PRINT

- Worksheet x24 (3 sheets each, double-sided, B&W)
- Spinner template, 160 gsm card x9 sheets (at 100%, includes 3 spares)
- Odd-one-out sheet x6 (1 A4 sheet each, cut into 4 A6 cards)
- Teacher copy: kit card (A5)

GET OR MAKE (NEW THIS LESSON)

- Paper clips x26 (28 mm, all the same size)
- Scissors x12 pairs
- Stopwatches x6
- Metre sticks x6
- Kit trays x6 (requisition lists 1: decide)
- A5 envelopes x6
- Masking tape x1 roll (Week 2 roll may do)
- Make: demo spinners x2 (S 4 cm, L 8 cm wings)
- Make: wall drop marks at 6 stations (2.0 m and 1.8 m)

Have ready from earlier weeks: Marker pen (Week 2), Pencils x24 (Week 1), Group role cards x6 sets, Classroom language cards x30, Sentence frame card sets x6, Language wall posters x4, Logbooks x24 in the class box, My Science Dictionary x28 (optional)